TEXORA · LECTURE SHEET

Comparison of Different Spinning Methods

Ring, rotor, compact, siro, air-jet, vortex and friction spinning all turn staple fibre into yarn — and no one of them is best at everything. This sheet sets the systems side by side on principle, count range, speed, structure, properties and economics, so a spinning method can be chosen from evidence rather than habit.

YM-4201 · MODERN YARN MANUFACTURING · TOPIC 3 · LEVEL 4 TERM 2 · YARN MANUFACTURING SPECIALIZATION

1. Principle recap — seven systems in seven sentences

- **Ring.** Roving is roller-drafted; twist and winding are performed together by a ring-and-traveller spindle rotating the package; true twist runs through the whole yarn.
- **Compact.** Ring spinning with the drafted strand pneumatically condensed before twist insertion, eliminating the spinning triangle; otherwise the same drafting, spindle and package.
- **Siro.** Two rovings are drafted together through one ring position; the strands twist together at a convergence point into a single yarn with a plied character.
- **Rotor.** Draw-frame sliver is opened to single fibres, reassembled in the groove of a high-speed rotor and twisted from the open end; winding is separate and packages are large.
- **Air-jet.** A drafted strand is held together by fringe fibres wrapped around a largely parallel, low-twist core by swirling air nozzles (false-twist principle).
- **Vortex (MVS).** A drafted bundle enters an air-vortex chamber around a hollow spindle; whirled fibre ends wrap a parallel core — a refined member of the air-jet family.
- **Friction (DREF).** Fibres are fed onto a fringe rolled between perforated suction drums; friction builds and twists the yarn, withdrawn along the drum axis.

Two divisions organise the comparison: **true-twist versus wrapping systems** (ring, compact, siro and rotor insert real twist; air-jet and vortex yarns are held together mainly by wrapping fibres), and **continuous-strand versus open-end systems** (ring-family systems never break the fibre stream; rotor and friction systems individualise and reassemble the fibres).

2. The large comparison table

All property figures are **typical textbook ranges** for comparable material and count, quoted so systems can be ranked and discussed — not specifications. Exact values must always come from the yarn actually tested. Tenacity is indexed to conventional ring yarn = 100 for the same fibre and a comparable count.

FEATURE	RING	COMPACT	SIRO	ROTOR	AIR-JET	VORTEX	FRICTION (DREF)
Typical count range	Widest: about 6 tex to over 200 tex (Ne about 3–100)	Medium to very fine; premium counts	Medium to fine	Coarse to medium, about 20–200 tex (Ne about 3–30)	Medium, about 12–40 tex; blends favoured	Medium, about 12–40 tex	Very coarse, about 100 tex and coarser; core yarns
Typical delivery speed	About 20–30 m/min	About 20–30 m/min	About 20–30 m/min	About 100–200 m/min	About 300–450 m/min	About 400–500 m/min	About 150–300 m/min, coarse counts
Twist insertion	Ring and traveller; spindle rotates package	As ring	As ring; strands self-ply	Rotor rotation; one revolution ≈ one turn	Air nozzles; false twist + wrapping	Air vortex; wrapping	Rolling between suction drums
Yarn structure	Helical, well oriented; surface hairs from the triangle	Helical, denser; almost all fibres incorporated	Two strands plied in one step; smooth, round	Core + sheath + wrapper fibres; bulky, less oriented	Parallel core, wrapped surface; lean, firm	Parallel core, spiral wrappers; clean surface	Rolled assembly, often core-sheath; bulky
Tenacity (ring = 100)	100 (reference)	About 105–115	About 105–115 vs equivalent singles	About 80–90	About 75–85	About 80–90	About 50–70

Hairiness	Highest of the quality systems	Lowest of the true-twist systems	Lower than singles ring yarn	Moderate-low; wrapper fibres present	Low	Very low	Moderate; less critical in its end uses
Evenness	Good; depends on roving	Good, as ring	Good; two rovings average each other	Very good short-term, from back-doubling	Good	Good	Adequate for coarse uses
Typical end uses	Everything, incl. finest shirting	Premium shirting, fine knits	Suiting, plied-character knits	Denim, towels, blankets	Polyester-rich apparel	Apparel, low-pilling fabrics	Blankets, mops, carpet backing
Raw-material tolerance	Needs good fibre for fine counts	As ring	As ring	Tolerates lower-grade, shorter cotton	Prefers synthetic, uniform fibres	Needs a uniform bundle	Highest: short, recycled fibres

The tenacity and delivery rows state the central trade: the fastest systems buy speed by giving up some of the helical, load-sharing structure that makes ring yarn strongest. Compact spinning is the deliberate exception — it keeps the structure and pays in machine cost and suction energy instead.

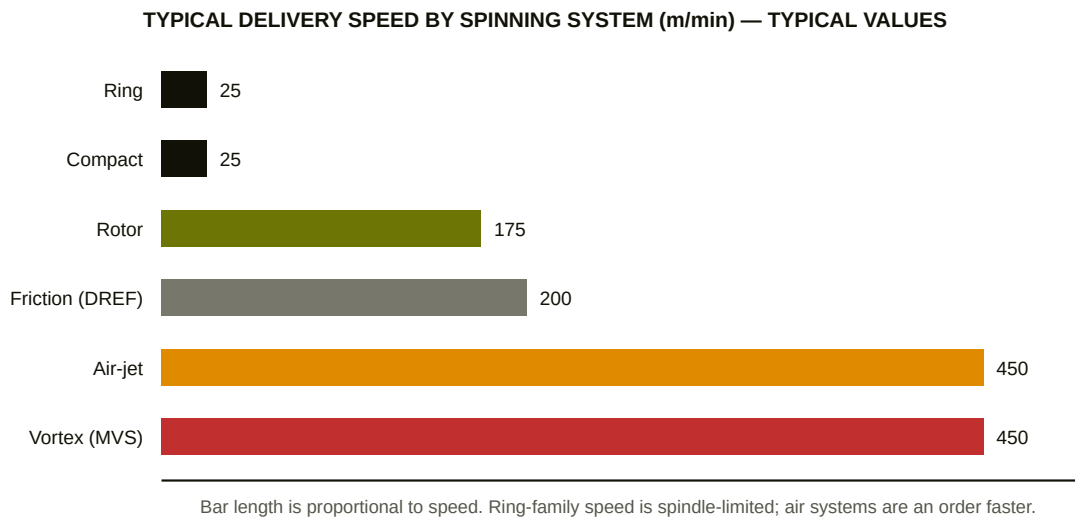


Figure 1 — Typical delivery speeds by system. Values are typical midpoints of textbook ranges (ring about 25, rotor about 150–200, air-jet/vortex about 400–500 m/min), shown for comparison, not as machine guarantees.

3. Reading the comparison — structure decides properties

Strength. Tenacity follows fibre orientation and load sharing. Ring and compact yarns put the highest fraction of fibres into a helix under tension, so they head the table. Rotor yarn loses strength to hooked fibres and non-load-bearing wrappers; wrapping systems lose it because a wrapped core cannot mobilise fibre strength like a twisted helix; friction yarn sits lowest — acceptably, because its end uses do not ask for fine-yarn strength.

Hairiness and evenness rank the systems differently. Vortex and compact sit at the clean extreme — one by wrapping the surface down, the other by never letting edge fibres escape — while ring yarn, strongest in structure, is hairiest in surface. Rotor mass variation is excellent because of back-doubling, even though fibre orientation is poor. Fabric trials, not single-yarn league tables, settle final selection.

Count range is a hard boundary. A very fine count cannot be rotor spun economically, because too few fibres would occupy the groove; a very coarse count is wasteful on a ring frame and natural on a DREF machine; 100% short cotton wraps badly in air-jet systems. Read the count-range row first in any selection exercise: it eliminates systems before economics is discussed.

4. Production economics

WORKED EXAMPLE — PRODUCTION PER POSITION AND PER ILLUSTRATIVE MACHINE, 20 TEX YARN

Production per position per hour = delivery (m/min) × 60 × tex / 1,000, in grams.

Ring at 25 m/min: 25 × 60 × 20 / 1,000 = **30.0 g/h**. At 95% efficiency: 28.5 g/h.

Rotor at 150 m/min: 150 × 60 × 20 / 1,000 = **180 g/h**. At 98% efficiency: 176.4 g/h.

Air-jet at 450 m/min: 450 × 60 × 20 / 1,000 = **540 g/h**. At 95% efficiency: 513 g/h.

Per position, air-jet makes 18 times the ring output and the rotor 6 times, at this count.

Per machine the picture narrows, because position counts differ. Illustrative frames — ring 1,200 spindles, rotor 600 rotors, air-jet 72 positions — at the efficiencies above give: ring 28.5 × 1,200 / 1,000 = **34.2 kg/h**; rotor 176.4 × 600 / 1,000 = **105.84 kg/h**; air-jet 513 × 72 / 1,000 = **36.94 kg/h**.

Position counts are illustrative, stated so the arithmetic can be followed; real frames vary by maker and generation.

Further effects: **stages saved** (rotor, air and friction lines delete roving and deliver dye-ready packages; compact deletes nothing and adds suction energy); **labour** (fewer operatives per kilogram on modern systems); **energy** (compressed air is expensive; rotor economics depend on the vacuum system); and **value, not only cost** (a compact yarn may justify higher spinning cost through weaving gains and a clearer fabric, while a cheaper rotor yarn wins where extra strength cannot be paid for). Ring frames

change count and material with least fuss; high-speed systems reward long runs. Most large mills run several systems side by side, routing each order to the system whose column fits it.

5. A selection procedure

1. Fix count and fibre first; strike out every system whose count range or fibre tolerance excludes them.
2. List the fabric properties that are genuinely critical — tenacity, hairiness, evenness — and rank the surviving systems on those rows only.
3. Cost the survivors per kilogram at realistic efficiency, including stages and energy.
4. Check run length and changeover pattern against each system's flexibility.
5. Confirm by spinning and fabric trial: table values are typical ranges, and the final evidence is the mill's own yarn in the customer's fabric.

Glossary

True twist — twist inserted permanently into the yarn structure. **Wrapping system** — a system holding a core together mainly by fibres wound around its surface. **Open-end system** — a system that individualises fibres before reforming them into yarn. **Siro yarn** — a yarn made by drafting two rovings together into one ring-spun strand with a self-plied structure. **Tenacity index** — tenacity expressed with conventional ring yarn as 100 for the same fibre and comparable count. **Delivery speed** — the rate, in m/min, at which a position delivers yarn. **Back-doubling** — fibre layering in a rotor groove that averages short-term variation. **Spinning triangle** — the unconsolidated fibre fan at ring twist insertion, removed in compact spinning. **Core-sheath yarn** — a yarn with a core of one material covered by a sheath of another.

Key points

• Ring = 100 is the tenacity reference; compact and siro exceed it slightly, rotor and vortex sit below it, friction yarn well below — all as typical ranges at comparable count and fibre.
• Delivery speed ranks oppositely: ring and compact about 25 m/min, rotor about 150–200, air-jet and vortex about 400–500.
• Count range and fibre tolerance are hard boundaries — apply them before any economic comparison.
• Hairiness, evenness and tenacity rank the systems differently, because each follows a different feature of yarn structure.
• Production per position = delivery × 60 × tex / 1,000 g/h; per machine, multiply by realistic position counts and efficiency.

• Stages saved, air and vacuum energy, labour and achievable selling value complete the economics — cost per kilogram alone does not choose a system.

Practice

1. For a 15 tex yarn, calculate production per position per hour (at 100% efficiency) for (a) a rotor position delivering 180 m/min and (b) a ring spindle delivering 20 m/min, and state the ratio of the two.
2. A fabric needs a very fine count, low hairiness and the highest available tenacity. Name the best-fitting system and give two reasons tied to rows of the table.
3. A mill must spin a very coarse yarn from recycled short fibre, including a core-sheath version. Name the system and explain which two characteristics make the others unsuitable by comparison.
4. Explain why a mill could rationally install compact spinning even though its delivery speed is no higher than conventional ring spinning.
5. Explain why tenacity figures in comparisons of spinning systems must state fibre and count.

Answer key

1. (a) Rotor: 180 × 60 × 15 / 1,000 = **162 g/h**. (b) Ring: 20 × 60 × 15 / 1,000 = **18.0 g/h**. Ratio = 162 / 18.0 = **9 times** the ring output per position.
2. Compact spinning. Its count range extends to very fine counts where rotor and air systems cannot operate satisfactorily, and its tenacity index (about 105–115 against ring = 100) with greatly reduced long hairiness matches both requirements directly.
3. Friction (DREF) spinning. It is built for very coarse counts at which fine-count systems cannot operate, and it has the highest raw-material tolerance, spinning short and recycled fibres — and feeding a core for core-sheath yarn — that would defeat rotor groove stability and air-jet wrapping.
4. Because value, not speed alone, decides: condensing removes the spinning triangle, so hairiness falls and tenacity rises at the same count, and downstream gains — weaving performance, clearer fabric, less singeing or sizing in suitable uses — can exceed the cost of the unit and its suction energy, especially on premium fine-count products.
5. Tenacity depends on fibre quality and on the number of fibres in the cross-section, which count controls. A figure quoted without them may compare a fine yarn of weak fibre with a coarse yarn of strong fibre and attribute the whole difference to the system — and mislead the selection.